

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

Branch: -AI/CPS/CS/DS

Semester: - III

Mid-Term Examination September-2026

Course : B.Tech

Subject Code: - CSE(AI/CPS/CS/DS)-3004

Max. Marks: 20

Subject Name: Formal Language and Automata Theory

Time: 90 Minutes

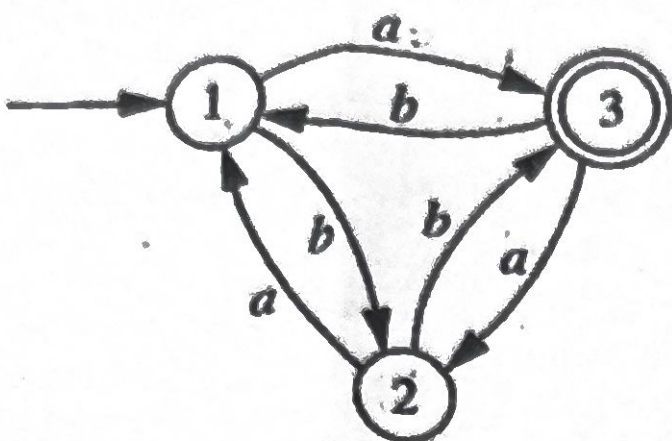
NOTE:

1. All the questions are compulsory.
2. While solving write question number and its sub part clearly.
3. Solve questions in given sequence and all the sub parts should be at same place.

1.A	Construct DFA that accepts all the string of {a, b} which do not end with "aba".	2
B.	Minimize the following DFA	3
2.A	Consider a Moore machine over input alphabet $\Sigma = \{0,1\}$ and output alphabet $\Delta = \{X, Y\}$ with the behavior The machine outputs X if the total number of 1's seen so far is divisible by 3; otherwise, it outputs Y.	2
B.,	Construct a regular expression corresponding to the language of all strings containing both 101 and 010 as substring over $\{0,1\}^*$	3
3.A	Find the Context Free Grammar for the following languages: (i) $L = \{a^l b^m c^n \mid m=l+n\}$ (ii) $L = \{a^n b^m \mid n \neq m \text{ where } n, m \geq 0\}$	2

Name of Student: _____

Scholar No.: _____

B.	Consider the DFA M shown below find regular expression accepted by this automaton  <pre> graph LR start(()) --> 1((1)) 1 -- a --> 2((2)) 2 -- b --> 1 1 -- b --> 3(((3))) 3 -- a --> 1 2 -- b --> 3 3 -- a --> 2 style start fill:none,stroke:none </pre>	3
4.A.	Find a FA that accepts the following Regular expressions: $((aa + bb + \epsilon)(ab + ba)^*)^* + a$	2
B.	Describe the Language generated by following Right Linear Regular Grammar $S \rightarrow aA / bC / b$ $A \rightarrow aS / bB$ $B \rightarrow aC / bA / a$ $C \rightarrow aB / bS$	3